Python

high level language → understandable by human

version-3

Python has simple syntax as English

Versitile → web page

Analytics ✓

Software

Statistics ✓ — R, SAS — licensed

machine learning } Distribution, values ↑%.

Deep learning }

Deep learning

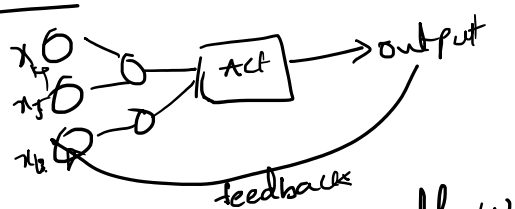
process → python Difficult → Beautyness of python

↓

Packages → 100's command.

Somebody written stored

→ syntax to process work



Python install website

SAS Inc → Dollers → 100 months 1 user

Anaconda ✓

import pandas

Python → use

commands to Execute Code.

1) ctrl + Enter → Execute only

2) shift + Enter → Execute as well as → insert a new cell.

goto next cell.

markdown cell

→ ESC + M.

Headings

My first program. → Big Size.

Today is my first class → little less

"(go to google")

Today is . . .

Hyperlink

click [this](https://www.google.co.in "Goto google")

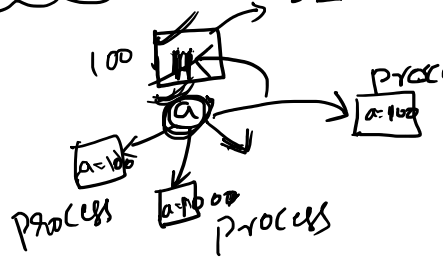
Variable:

storing place of values

✓ Numeric < int → 24.3
Float → decimal
Bool → True → 1, False → 0

✓ Character

& modify the value.



✓ a1
✓ a2
✓ a3

✗ 1a
1b
1c

Rules to create variable

1) should contain alphabets, numbers and underscore
2) you cannot start variable name with number
3) it should not be keyword python. (If, for)
4) it should not have any space btw name.

NumVar = 10 ✗
Num_Var = 10 ✓

datatype

1) Integer. 24.3
2) Float. 24.3
3) boolean 1/0. True / False.
4) character/string 'strings'
5) date '01/7/2025'

to check datatype of a variable
⇒ type (variable name)

assigning / creating multiple variable

a = 10
b = 20
c = 30

a, b, c = 10, 20, 30

assigning a single value to multiple variable

a = b = c = 100

Code → Execute.

Mark down → code

python comments

1) single line comment → #
2) multiple line comments → ''' ... '''

shortcut key for comments in JP.

ctr + ?

Identifier

Variable a = 100, a = 2

vs

Indentifier

2) Inbuilt Indetifiers

2) user define Indetifier

total Amount ✗

total = program

total = a + b ✗

total = 200

keywords ✗

assign literals

wrongly direct

Kegw.

import keyword

print (keyword. kwlist)

Type convertion

int, float, bool, str.

int, float, bool, str.

24.3 → float
↓
int.

24.

Arthmatic operators

+, −, *, /, //, **, %

add, sub, mul, div, int div, power, remainder

a = 3
b = 2

3 ** 2 = 3

$3^2 \Rightarrow 3 \times 3 = 9$

3.75

25 % 7 → 4

% ⇒ 7) 25 (3.
21
4

21.68
25/7 =

25 // 7 ⇒ 21

25 / 7 = 21.67

25 % 7 = 4

Comparison operators → True/False

>, <, ==, !=, >=, <=

x = 20
y = 15.

x > y → bool True

x < y = False.

x == y ⇒ False.

x != y → True

x >= y → True

x <= y → False.